

EXPERIMENT 6

PORT SCANNING USING ANGRY IP SCANNER

Aim

To familiarize students with Angry IP Scanner and use it to discover active devices, identify host information and check open ports within an authorized IP range.

Description

Angry IP Scanner is an open-source IP address and port scanner. It provides a graphical interface for scanning a range of IP addresses and identifying active or reachable devices.

The tool can collect and display information such as IP address, hostname, ping, packet loss, MAC address, MAC vendor, open ports, TTL and HTTP information. It is useful for basic network discovery and identifying active devices.

Requirements

Category	Requirements
Hardware	Computer or laptop; active network connection.
Windows	Windows operating system; Angry IP Scanner; network connection.
Kali Linux	Kali Linux; Angry IP Scanner package; network connection.
Other	Authorized IP range and permission to scan the selected network.

Procedure

A. WINDOWS PROCEDURE

Step 1 — Install Angry IP Scanner

Download and install the Windows version of Angry IP Scanner. After installation, launch the application.

Step 2 — Launch Application

Open Angry IP Scanner from the Windows Start menu or installed shortcut.

Step 3 — Enter the IP Range

Enter the authorized IP range that needs to be scanned.

Example

Start IP: 192.168.1.1

End IP: 192.168.1.254

Step 4 — Configure Fetchers

Select the information to be collected: Ping, Hostname, Ports, MAC Address, MAC Vendor, TTL and HTTP.

Step 5 — Start the Scan

After entering the IP range and configuring the required fetchers, click Start. The scanner checks the specified IP addresses and displays the results.

Step 6 — Analyze Results

Observe active hosts, IP addresses, hostnames, ping responses, open ports and MAC addresses. The available information depends on the host and selected scanning options.

Step 7 — Export Results

The scan results can be exported for documentation and further analysis.

B. KALI LINUX PROCEDURE

Step 1 — Download the Package

Download the appropriate Angry IP Scanner package for Kali Linux. For a 64-bit Debian-based Kali installation, the .deb package can be used.

Step 2 — Install the Package

Open the Kali Linux Terminal and navigate to the directory containing the downloaded package. Install it using:

Commands

```
sudo dpkg -i ipscan-<version>-amd64.deb
```

```
If dependency problems occur:  
sudo apt -f install
```

Step 3 — Launch Angry IP Scanner

After installation, launch the application using:

Command

```
ipscan
```

Step 4 — Enter the IP Range

Specify the authorized range to be scanned. For example:

Start IP: 192.168.1.1

End IP: 192.168.1.254

Step 5 — Configure Fetchers

Select Ping, Hostname, Ports, MAC Address, MAC Vendor, TTL and HTTP according to the information required.

Step 6 — Start Scanning

Click Start to begin scanning. Angry IP Scanner checks the IP addresses in the selected range and displays the discovered information.

Step 7 — Analyze the Results

Examine the results for active devices, IP addresses, hostnames, ping responses, available ports, MAC addresses and other enabled information.

Result Analysis

Angry IP Scanner successfully scans the selected IP range and identifies reachable devices. The scanner displays information such as the IP address, hostname and ping response for discovered devices.

When the corresponding fetchers are enabled and the information is available, additional details such as MAC address, MAC vendor and open ports can also be displayed.

IP address analysis identifies each device detected within the scanned range. Hostname information, when available, provides a more recognizable identification of the host.

Ping information provides an indication of host reachability. MAC address and MAC vendor information can provide additional details about a network interface and its associated manufacturer when available.

Selected ports can be checked during the scan to identify network services that may be available through those ports. The graphical interface makes it easier to examine a large IP range without manually entering individual commands for every address.

The scan results can be exported so that the discovered information can be retained for documentation and later analysis.

Result

The specified IP range was successfully scanned using Angry IP Scanner, and active hosts and available network information were identified.

Observations

1. Angry IP Scanner is a graphical network scanner.
2. It can scan a specified range of IP addresses.
3. Active and inactive hosts can be identified.
4. Ping responses can be observed for discovered hosts.
5. Hostname information can be displayed when available.
6. MAC addresses may be obtained when supported.
7. MAC vendor information can be displayed.
8. Open ports can be identified according to the selected scanning configuration.
9. Scanning results can be exported for further analysis.
10. The tool is available for both Windows and Linux.
11. Kali Linux can launch Angry IP Scanner using the `ipscan` command.
12. The scanning range can be specified using starting and ending IP addresses.
13. Different fetchers can be enabled according to the information required.
14. Ping, hostname, ports, MAC address, MAC vendor, TTL and HTTP information can be selected.
15. The graphical interface makes it easier to analyze a large IP range.
16. The scanner can help identify reachable devices within an authorized network.
17. The information displayed depends on the selected scanning options and information available from the target.
18. Scan results can be retained by exporting them for documentation.
19. Angry IP Scanner can be useful for basic network discovery and security auditing.
20. Scanning should only be performed on authorized networks and systems.

Practical Flow: Install → Enter IP range → Select scanning preferences → Start scan → Inspect hosts and ports → Save results.